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Jawaban

1 Daerah IR dengan luas

$$\sum_{i=1}^{\infty} \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$$

Konversi $\int \rightarrow \sum$

$$\Rightarrow \int_a^b f(x) dx = \sum_{i=1}^{\infty} f(x) \Delta x$$

$$= \sum_{i=1}^{\infty} \underbrace{f\left(a + \frac{(b-a)i}{n}\right)}_{f(x)} \underbrace{\left(\frac{(b-a)}{n}\right)}_{\Delta x}$$

Maka diketahui

$$a=1 \Rightarrow b-a=3 \Rightarrow b=4 \text{ dan } f(x) = \sqrt{x}$$

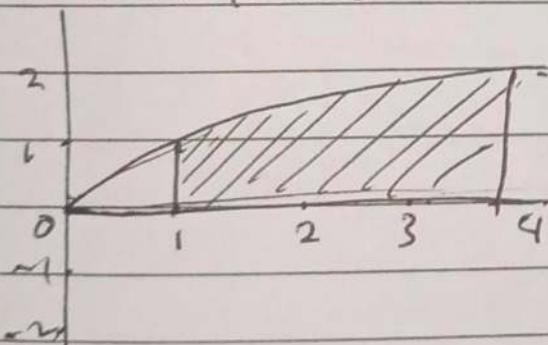
Sehingga

$$\sum_{i=1}^{\infty} \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$$

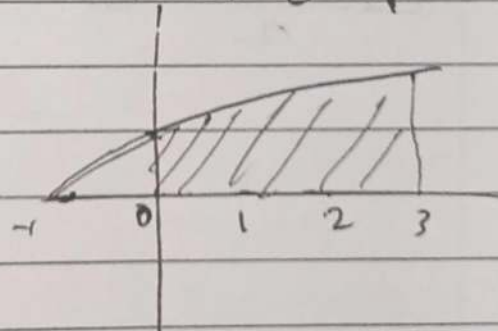
$$\int_1^4 \sqrt{x} dx = \frac{14}{3}$$

$$\int_0^3 \sqrt{1+x} dx$$

Sketsa kasar



Basis 0 (pauze)



2. Pers Singgung kurva $y = g(x)$; $x^2 = -1$

$$g(x) = \int x^2 \sec(t-1) dt + 3$$

Berarti $m = g'(c)$ dengan $c = -1$

$$\frac{d}{dx} g(x) = \frac{d}{dx} \int_1^x \sec(t-1) dt + \frac{d}{dx} 3$$

$$= \frac{d}{dx} \left(F(x^2) - F(1) \right) + 0$$

$$= 2x \cdot \sec(x^2-1)$$

$$\text{Saat } x = -1 \Rightarrow g'(-1) = -2 \cdot \sec(0) = -2$$

$$y = -2x + C$$

Agar bersinggungan, $g(-1) = y(-1)$

$$\Rightarrow g(x) = \int_{-1}^{x^2} \sec(t-1) dt + 3 = 3$$

$$\Rightarrow 3 = 2 + C$$

$$\Rightarrow C = 1$$

Dengan demikian

$$\boxed{y = -2x + 1}$$

3 (a) $\int_{-\pi}^{\pi} \frac{\cos t}{\sqrt{4+3\sin t}} dt$

Substitusi $u = 4 + 3\sin t$

$$\Rightarrow du = 3 \cos t dt$$

Evaluasi batas, $x \in \{-\pi, \pi\}$

$$x = -\pi \Rightarrow u = 4 + 3\sin(-\pi) = 4$$

$$= \pi \Rightarrow u = 4 + 3\sin(\pi) = 4$$

Bentuknya,

$$\int_4^4 g(u) du = \boxed{0} \text{ karena } \int_a^a f(x) dx = 0$$

(b) $\int_1^9 (2\sqrt{y}(1+\sqrt{y})^2)^{-1} dy$

$$= \int_1^9 \frac{1}{2\sqrt{y}(1+\sqrt{y})^2} dy$$

Misalkan $u = 1 + \sqrt{y} \Rightarrow u_0 = 2, u_1 = 3$

$$\Rightarrow du = \frac{1}{2\sqrt{y}} dy$$

$$\Rightarrow \int_2^3 \frac{1}{u^2} du = -\frac{1}{u} \Big|_2^3$$

$$= \frac{1}{2} - \frac{1}{3}$$

$$= \boxed{\frac{1}{6}}$$

$$(c) \int_{-1}^1 \frac{|x^3| + x^3}{f(x)} dx$$

Pecah, untuk $x \geq 0$, $f(x) = x^3 + x^3 = 2x^3$

$$x < 0, f(x) = (-x^3) + x^3 = 0$$

Maka, ekuivalen dengan

$$\int_0^1 2x^3 dx$$

$$= \frac{1}{2} x^4 \Big|_0^1$$

$$= \frac{1}{2}$$

$$H(t) = \sqrt{t+1} + 5t^{\frac{1}{3}}, \quad t \in [0, 8]$$

$$(a) \text{ Rata-rata} = \frac{\int_0^8 H(t) \cdot dt}{8-0}$$

$$= \frac{1}{8} \int_0^8 \sqrt{t+1} + 5\sqrt[3]{t} \, dt$$

$$= \frac{1}{8} \cdot \left[\frac{2}{3} \sqrt{(t+1)^3} + \frac{3}{4} \cdot 5 \sqrt[3]{t^4} \right]_0^8$$

$$= \frac{1}{8} \cdot \left(\frac{2}{3} \cdot \sqrt{9^3} + \frac{3}{4} \cdot 5 \cdot 2^4 \right) - \frac{2}{3} \cdot \frac{1}{8}$$

$$= \frac{1}{8} \cdot \left(78 - \frac{2}{3} \right)$$

$$= \frac{29}{3} = 9.666$$

(b) Menurut Teorema Nilai Rata-rata untuk Integral, pasti ada nilainya,

$$\sqrt{t+1} + 5t^{\frac{1}{3}} = \frac{29}{3}$$

Tentu sulit untuk mencari pakai persamaan aljabar karena akar-akarnya cukup ribet sehingga digunakan metode analitik dengan meninjau bahwa fungsi tersebut monoton naik $t \in (0, 8)$, kemudian pakai binary search ↗
varis monoton

ambil $t = \frac{8+0}{2} = 4$, $H(4) \approx 10,17 > 9,66 \Rightarrow \text{maks} = 4$

$t = \frac{4+0}{2} = 2$, $H(2) \approx 8,03 < 9,66 \Rightarrow \text{min} = 2$

$t = \frac{4+2}{2} = 3$, $H(3) \approx 9,21 < 9,66 \Rightarrow \text{min} = 3$

$t = \frac{3+3}{2} = 3,5$, $H(3,5) \approx 9,71 > 9,66 \Rightarrow \text{maks} = 3,5$

$t = \frac{3,5+3}{2} = 3,25$, $H(3,25) \approx 9,96 < 9,66 \Rightarrow \text{min} = 3,25$

$t = \frac{3,5+3,25}{2} = 3,375$, $H(3,375) \approx 9,59 < 9,66 \Rightarrow \text{min} = 3,375$

$t = \frac{3,375+3,5}{2} = 3,4375$, $H(3,4375) \approx 9,65 < 9,66$, $\text{min} = 3,4375$

$t = \frac{3,4375+3,5}{2} = 3,46875$, $H(3,46875) \approx 9,68 > 9,66$, $\text{maks} = 3,46875$

Maka $t \approx \frac{3,46875 + 3,4375}{2}$

$= \boxed{3,453125}$

☐	5.	x	1	1.5	2	2.5	3	3.5	4	4.5	5
☐		$f(x)$	2.4	2.9	3.3	3.6	3.8	4.0	4.1	3.9	3.8

☐ (a) Metode ujung kiri/kanan bisa memakai $\Delta x = 0.5$ karena point/titik-titiknya terdefinisi, sedangkan midpoint menggunakan $\Delta x = 1$

$$[1, 2] \rightarrow 1.5 \Rightarrow 2.9$$

$$[2, 3] \rightarrow 2.5 \Rightarrow 3.6$$

$$[3, 4] \rightarrow 3.5 \Rightarrow 4.0$$

$$[4, 5] \rightarrow 4.5 \Rightarrow 3.9$$

$$\text{Sehingga } A = \sum f(x) \Delta x$$

$$= (f(1.5) + f(2.5) + f(3.5) + f(4.5)) \cdot 1$$

$$= 2.9 + 3.6 + 4.0 + 3.9$$

$$= \boxed{14.4}$$

☐ (b) Menggunakan deret Taylor (Calc II)

$$f(x) \approx f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2$$

Integrasikan

$$\int_a^b f(x) dx \approx \int_a^b f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 dx$$

$$\Rightarrow \int_{c-\Delta x}^{c+\Delta x} f(x) dx \approx \int_{c-\Delta x}^{c+\Delta x} f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 dx$$

gaji terhadap $x-c$

$$= f(c) \Delta x + 0 + \frac{f''(c)}{6} (x-c)^3 \Big|_{c-\Delta x}^{c+\Delta x}$$

$$= f(c) \Delta x + \frac{f''(c)}{24} \Delta x^3 \quad (1)$$

$$e = A_{\text{akt}} - A_{\text{aproksimasi}}$$

$$= (1) - f(c) \Delta x$$

$$= \frac{f''(c)}{24} \Delta x^3$$

Total e (ada n)

$$= n \cdot \sum_{i=1}^n \frac{f''(c_i)}{24} \Delta x^3$$

$$< \frac{n \cdot K \cdot \Delta x^3}{24}$$

dengan K maks. nilai $f''(x)$

Maka,

$$|E| \leq n K \Delta x^3$$

$$= \frac{n \cdot 3 \cdot \left(\frac{4}{n}\right)^3}{24}$$

$$= \frac{4 \cdot 3}{24} = \boxed{\frac{1}{2}}$$

Berarti, $|A_{\text{aktual}} - A_{\text{aproksimasi}}| \leq \frac{1}{2}$

6. $\int_0^1 (2x+3)^{-1} dx$ dengan parabola/Simpson, $|E| < 10^{-3}$

$$f(x) = (2x+3)^{-1}$$

$$f'(x) = -2(2x+3)^{-2}$$

$$f''(x) = 8(2x+3)^{-3}$$

$$f'''(x) = -48(2x+3)^{-4}$$

$$f^{(4)}(x) = 384(2x+3)^{-5}, \text{ monoton turun saat } x \geq 0, x \in (0,1)$$

Sehingga $\max f^{(4)}(x), x \geq 0$

$$\Rightarrow \max |f^{(4)}(x)|, x \in [0,1] = \frac{384}{243} = \frac{128}{81}$$

$$|E| \leq \frac{(\max |f^{(4)}(x)|) \cdot \Delta x^5}{180n^4}$$

$$= \frac{128}{81} \cdot \frac{1}{n^4} < \frac{1}{1000}$$

$$\Rightarrow n^4 > \frac{1280}{180} \approx 7.11$$

$$\Rightarrow n > \sqrt[4]{7.11} \approx 1.72$$

Maka, ambil $n=2$

Ini berarti ada 1 parabola melewati titik

$$\{0, \frac{1}{2}, 1\} \Rightarrow f(0) = \frac{1}{3}$$

$$f(\frac{1}{2}) = \frac{1}{4}$$

$$f(1) = \frac{1}{5}$$

$$A = \frac{1}{3} (f(0) + 4f(\frac{1}{2}) + f(1)) \cdot \Delta x$$

$$= \frac{1}{3} \left(\frac{1}{3} + \frac{1}{4} \cdot 4 + \frac{1}{5} \right) \cdot \frac{1}{2}$$

$$= \frac{1}{6} \left(\frac{18}{15} \right)$$

$$= \frac{23}{90} = 0.2555$$

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